From: **Transactions on Interactive Intelligent Systems** onbehalfof@manuscriptcentral.com
Subject: [ACM TiiS] Editorial decision for TiiS-2026-Feb-022, "From Experts to Language Models: Enhancing Online Expert Elicitation with LLMs"
Date: May 8, 2026 at 00:40
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07-May-2026

Dear Ms. Ngirwa,

Based on the recommendation of the Associate Editor (AE), I'd like to inform you that your submission is "Rejected with option to resubmit after rewriting".

As you will see below, the reviewers and the AE did not find that the manuscript could be accepted at this point, even conditionally, in line with the rigorous standards of ACM journals. That said, they did encourage a resubmission to TiiS, assuming that the new version effectively deals with the issues identified by the reviewers and the AE.

I'd like to thank you for your submission and hope that you submit a new version to TiiS. Look forward to seeing your revised manuscript.

Best regards,
Shlomo Berkovsky
ACM TiiS editor-in-chief

DECISION CATEGORY:

"Rejected with option to resubmit after rewriting"

EXPLANATION OF DECISION CATEGORY (from http://tiis.acm.org/reviewing-procedure.html#categories): "The authors are encouraged to submit a rewritten version of the manuscript, taking into account the critiques and suggestions of the reviewers and the associate editor. No specific conditions for acceptance of the rewritten submission are listed. If such a submission is made, it will be reviewed in its entirety. The set of reviewers will usually include some or all of the original reviewers."

ASSOCIATE EDITOR/GUEST EDITOR COMMENTS:

Associate Editor
Comments to the Author:
The paper presents timely studies on using LLMs to approximate human expert judgments in structured expert elicitation tasks. It also offers some novelty.
However, the paper has several major drawbacks as highlighted by the reviewers. The paper lacks details in its presentation. The experiment design needs to be better described. Several methodological choices need to be more clearly reframed as limitations. The statistical analysis ought to be more rigorous.
The proposed changes are numerous and require a good amount of time. So having 90 days to revise the paper seems adequate.
As noted by one reviewer, the authors should discuss their choice of treating the experts' answer as accurate and compare them with LLMs.

REVIEWS (including author clarification, if submitted; if there has been any change relative to the reviews that were sent for clarification, it will have been mentioned in the editors' comments above):

Reviewer: 1

Comments to the Author
This paper investigates whether large language models (LLMs) can approximate human expert judgments in structured expert elicitation (EE) tasks, using DC energy systems as a case study. The authors build on a group-based EE study with 24 domain experts and compare these expert responses to outputs from four open-source LLMs across three simulation conditions: zero-shot prompting, context-enriched prompting (based on the same background information provided to experts), and anchored prompting. The evaluation focuses on three types of elicitation questions—feasibility, importance, and barrier selection.

Major Concerns:

Overall contribution and generalizability:
The contribution is not clearly articulated and appears incremental relative to existing work on LLMs in expert-like tasks.
Claims about LLMs as "co-experts" or substitutes for expert elicitation are not sufficiently supported by the study design or results.
The generalizability discussion is too limited given the narrow setup (single domain, small expert sample, selected models). The authors should clearly reflect how these results can actually be operationalized or transferred to other domains. It stays questionable what researchers and practitioners can do with the results. The authors should cleraly outline the implications of their work.

Clarity and Readability:
The manuscript (especially the introduction) lacks a clear thread: the problem, methodology, and contribution are not coherently linked, making it difficult to understand the study setup early on.
The research questions are poorly formulated (especially RQ2 and RQ3), with grammatical issues and unclear intent.
Overall writing quality is inconsistent, with unclear phrasing and several sentences that reduce readability and precision.
One example:
„In fact, one of the key motivations for this work was to enable asynchronous group-based expert elicitation, allowing experts to contribute without being online at the same time." —> that was never clear throughout and the authors never introduced that

Methodological rigor:
The experimental design is not sufficiently transparent: it remains unclear how expert data (group-based sessions) is made comparable to LLM outputs - is it enough to use a limited number of experts? How consistent were the experts?

The construction of the "context" condition (based on post-processed transcripts via ChatGPT) is not well described or justified, raising concerns about reproducibility.
The post-processing of LLM outputs (e.g., removing duplicates and dropping iterations) is questionable and not sufficiently justified, potentially biasing results.
How would the results change with different models? What is essentially the impact of the temperature T?

Results and analyses:
The statistical analysis lacks rigor: CLMM is insufficiently reported (no assumptions, no clear model specification, no correction for multiple testing).
The interpretation of results is problematic: claims about "overconfidence" and "over-extremity" are overclaimed and rely on weak proxies.

Discussion:
The discussion remains largely descriptive and does not sufficiently engage with existing literature on LLM behavior, robustness, and evaluation.
Key factors such as temperature, model differences, and prompt sensitivity are not analyzed in depth.
Several interpretations are overstated relative to the evidence.
The authors should also discuss their findings with other related fields: there are so many studies that compare expert behavior with LLMs. Here is one examples:
Singhal, K., Azizi, S., Tu, T., Mahdavi, S. S., Wei, J., Chung, H. W., ... & Natarajan, V. (2023). Large language models encode clinical knowledge. Nature, 620(7972), 172-180.

Minor Concerns
RQ2 is not grammatically correct; RQ3 is also unclear and should be reformulated.
Ethical clearance is only briefly mentioned and lacks detail (e.g., approving body, scope, consent procedure).
It is unclear how consistent expert responses are and whether agreement was assessed.
Some terms and statements are vague or unclear (e.g., "design of EE systems that collaborate with Human-LLMs").
The role of asynchronous expert elicitation is introduced late and not integrated into the main narrative.
Results are sometimes presented descriptively without sufficient explanation (e.g., why certain models rate consistently higher).
Several grammar issues and incomplete or awkward sentences throughout the paper.

Reviewer: 2

Comments to the Author
This paper demonstrates that LLMs (i.e., Gemma3:12B (Gemma3:12B),
LlaMa-Pro (LlaMa-Pro), Mistral (Mistral), and Phi4 (Phi4)) can approximate structured expert judgment, that giving them more context generally improves alignment, and that they exhibit human-like cognitive biases when exposed to leading or anchoring information. This paper adds to the literature or discussion on whether LLMs can be good enough to replicate human expert judgment. I'm providing feedback below as well as concrete revision suggestions.

Missing context in EE related work (esp. in DC technologies).
1. The related work in EE needs a bit more work. I acknowledge that I am not an expert in Expert Elicitation, so the current sec 2.1 "The Expert Elicitation" section does not quite show the gap that this work attempts to fill in the literature. Also, this term sounds a lot like the Delphi method, which I encourage the authors to mention in the literature. Since this paper focuses on "Direct Current (DC) technologies" expert, I think that the authors could describe how EE in specifically DC technology work without LLMs so that the audience could have some context on the gap of what the authors are proposing to fill.

2. Unclear instructions to LLMs. I understand that the authors instructed both experts and LLMs questions in Table 3, but I think I didn't get the full context of the instruction. The appendix included different prompt templates, but it seems that the prompts just focus on this one project called "Shift2DC"? The exact question/prompt template requires further clarification, without which is hard to gauge the feasibility of the task itself (both for human experts and LLMs).

Unclear Significance of the Results
Perhaps related to the missing context above, the results section (and the RQs) made me feel that the authors are trying to show that LLMs (specifically the four models) and expert and human experts are "aligned" in this DC technology context. However, the findings

(1) by themselves perhaps are not so surprising now (understandably when the study was conducted back in 2024), and this further shows that this work needs strong motivation in the related work in DC EE field and how similar LLM simulation studies in other fields cannot generalize to DC EE field (e.g., perhaps mention the unique challenges here).

(2) lack validity check. Throughout the paper, the authors treat expert responses as a reliable reference standard and measure LLM performance in terms of how closely model outputs resemble that of the EE. But are experts' answer accurate? For instance, for the question (e.g., "How feasible is the use of DC in the previously described demonstrators?"), there seems to be a ground-truth question available. It would be fascinating to show that experts can be right most of the time for some projects (or vice visa). I think that the authors need to both acknowledge this limitation and discuss why "alignment" (and/or "correctness") means for both LLMs and human experts in DC technologies.

MISC:
1. l53 "play" --> "plays"
2. l81 sentence is incomplete
3. l284: "LLMs does not work" --> "LLMs do not"
4. l890: "variability in LLMs's responses." --> "variability in LLMs' responses."
5. The model names are sometimes inconsistent, but should be a quick change.

Reviewer: 3

Comments to the Author
This is an ambitious manuscript. The experimental matrix is substantial and the anchor-sensitivity analysis in Section 4.3, especially, is novel and distinctive. However, the current framing of LLMs as emulating, replicating, or serving as "functional analogs" of expert reasoning seems overstated. Matching expert response distributions on five-point ordinal items does tell us about output distributions on a structured task but not about how models weigh evidence or hold causal beliefs.

Recasting the contribution around distributional alignment with expert elicitation responses and prompt-induced shifts in LLM elicitation outputs would be both more precise and more interesting given your anchor results.

Two methodological choices should be reframed as limitations. First, you exclude expert discussion "to ensure a smooth blend with LLMs

Two methodological choices should be reframed as limitations. First, you exclude expert discussion "to ensure a smooth blend with LLMs simulation-generated responses" -- but the discussion is the elicitation in any group-based protocol. This is where calibration, disagreement, and revision happen, so excluding it makes the human side of the comparison closer to a survey than to elicitation. I would like to see a supplementary analysis using pre- and post-discussion ratings or qualitative coding of the transcripts.

Second, from what I can tell, the "context" condition is not the context experts received: it is Read.AI text post-edited in ChatGPT-4.1 and lightly paraphrased before being passed to the open-source LLMs. Things like salience structure, anchoring potential, and stylistic register of the input have all been touched by a frontier model. It would be useful to share the verbatim original transcript and the GPT-4.1 output in the supplementary materials, as well as to quantify how much the editing step changed. This can be discussed as a limitation in the main text rather than only in the AI-use declaration.

Section 5.3 sketches several promising directions but leaves the paper closer to a domain-specific benchmarking study than to an interactive-systems contribution. Consider committing to one or two concrete design roles your evidence supports, e.g. LLMs as anchor-sensitivity probes that flag bias-prone items before expensive human sessions. This would treat LLMs as diagnostic instruments for elicitation design rather than as synthetic participants (a framing that has serious empirical and ethical problems in the recent literature).

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According to general TiiS policy, after a decision of "Rejected with option to resubmit after rewriting", a resubmission is to be submitted within 90 days. The exact date - which may be adjusted on the basis of correspondence between the you and the editor-in-chief - will be found in your Author Center in the ScholarOne Manuscripts system (http://mc.manuscriptcentral.com/tiis).

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